

Roll No.

TPE-501

B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER) MID SEMESTER EXAMINATION, Oct., 2023

PETROLEUM GEOPHYSICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What do you understand by Geophysics ? How it plays a major role in Oil & Gas Industry ? What are the various earth properties measured in Geophysics ? (CO1)

OR

- (b) What are the various earth properties measured by Geophysical method ? How are these properties used in defining the subsurface structure ? (CO1)
2. (a) How many geophysical methods are available for oil and exploration ? Explain the Gravity survey method in details. (CO2)

OR

- (b) Explain passive and active method of geophysics. What is the objective of geophysical investigations ? (CO2)

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3. (a) What is Magnetic survey method ? Explain in detail. Name the natural magnetic elements and Ferromagnetic minerals. (CO2)

OR

- (b) How Electromagnetic methods apply for geophysical survey ? Draw a neat diagram to explain it. Where is this survey useful ? (CO2)
4. (a) What is the seismic method for exploration ? What are the basic properties measured in this method ? How is the seismic trace generated ? (CO1/CO2)

OR

- (b) Explain P-wave and S-wave along with figure. Write the formula for P-wave and S-wave velocity in terms of bulk modulus and shear modulus. (CO1/CO2)
5. (a) Explain Huygens' principle for wave propagation. Draw a diagram for wave propagation and refraction. (CO2)

OR

- (b) Explain Snell's law of reflection and refraction. What is acoustic impedance and reflection coefficients ? Write their formulas. (CO2)

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**B. TECH. PETROLEUM ENGINEERING (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
WELL LOGGING & FORMATION EVALUATION**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Write the principle of measurement in well logging technology, what are the reservoir parameters measured by logging tools in the well ?

(CO1)

OR

- (b) Write the history of the various generation logging tools and techniques adopted in the industry time to time. What is the latest logging technology now a days ?

(CO1)

2. (a) Write the physical properties of the sedimentary rocks ? What are the basic logs for estimation of reservoir parameters ?

(CO2)

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OR

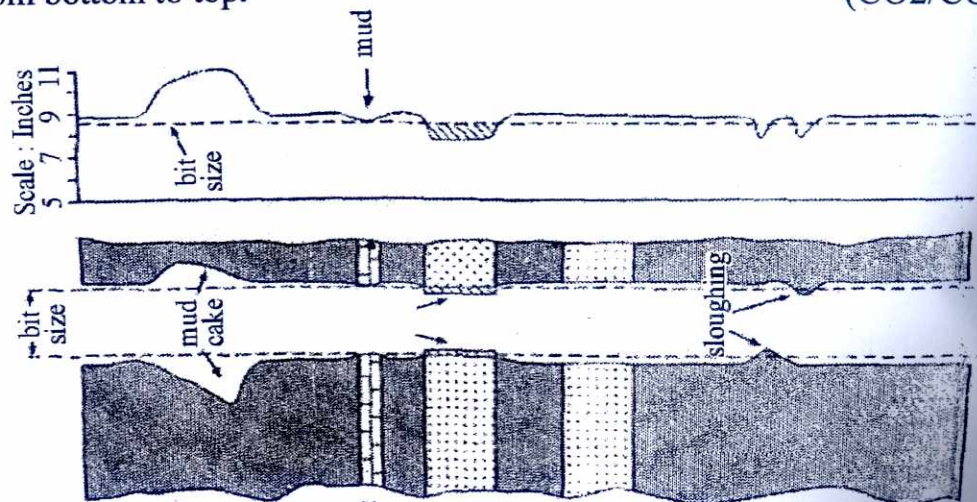
- (b) Explain Primary, Secondary porosity, Effective and Total porosity. What are the various parameters responsible for reduction of porosity explain with figure? (CO2)
3. (a) Draw a graph showing relative abundance Gamma ray emission is produced by three radioactive series found in the Earth's crust. (CO2)

OR

- (b) Write formula for volume of shale estimation using GR Log. Calculate V_{shale} if GR Log = 45, GR Shale = 120 & GR Clean 30. (CO2)
4. (a) What is SP log and how it is generated? What are the various types of potential responsible for generating SP log? Explain with figure of ionic movement. (CO1/CO2)

OR

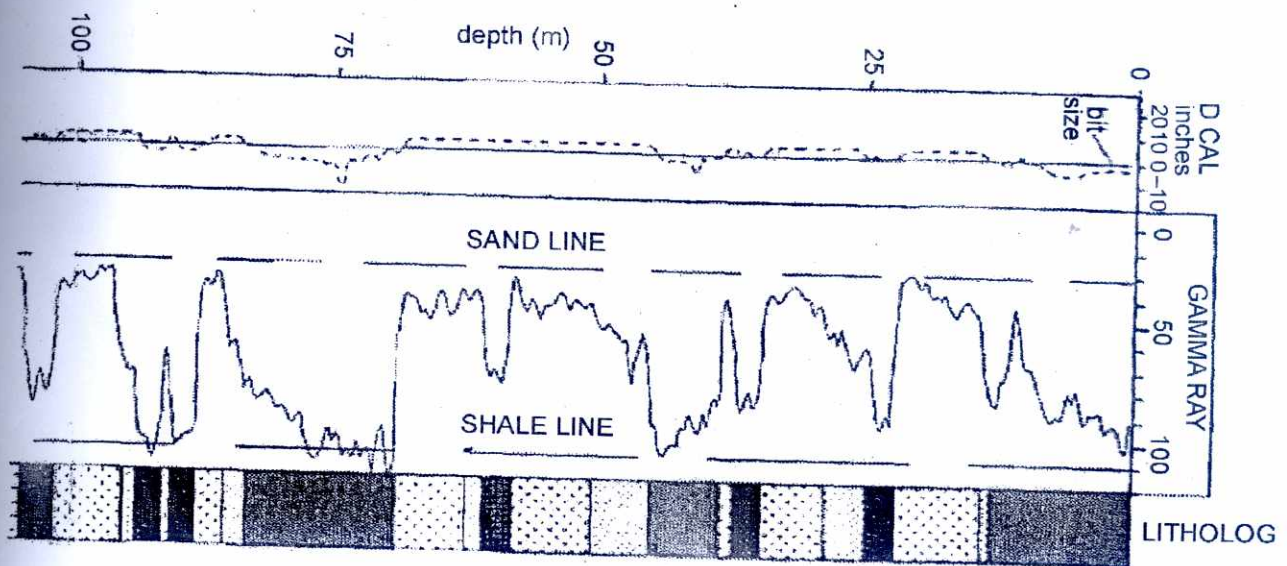
- (b) Explain resistance and resistivity. Write the various zones created during the invasion process in the borehole with figure. (CO1/CO2)
5. (a) What is the principle of caliper tool? Write the interpretation of the different zones given herewith in the figure based on caliper reading from bottom to top. (CO2/CO3)



(3)

OR

- (b) How GR log will be used for identification of different type of lithologies with depth ? Explain from this figure. (CO₂/CO₃)



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B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
PETROLEUM PRODUCTION ENGINEERING-I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Describe dehydration and process of dehydration. (CO2)

OR

- (b) With a neat and clean diagram explain the concept of vertical and horizontal three- phase separator. (CO2)

2. (a) Explain oil demulsification and process of demulsification. (CO1/CO2)

OR

- (b) With the help of a figure explain the function of Oil and Gas Facility. (CO1/CO2)

3. (a) Explain Emulsion and types of emulsion. (CO1/CO2)

OR

- (b) Describe Early Production Facility and GGS with function of GGS. (CO1/CO2)

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4. (a) Describe gas sweetening, types of amines used for gas treating, process of gas treating. (CO1/CO2)

OR

- (b) Describe Surface and Subsurface Production Equipment. (CO/CO2)

5. (a) Write short notes on the following : (CO1)

(i) Packers

(ii) Safety Valves

(iii) Gathering Pipeline

(iv) Transmission Pipeline

OR

- (b) (i) Describe Bridge Plugs, Safety Valves and Packers. (CO1)
- (ii) With the help of a table of weight percentage explain composition of oil and gas.

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**B. TECH. (PETROLEUM ENGINEERING)
(FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
RESERVOIR ENGINEERING—II**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) With the help of a graph explain the concept of Constant Terminal Rate Solution for semi-steady state non-Darcy flow. Explain with the help of Darcy's law for radial flow. (CO1)

OR

- (b) Illustrate dimensionless variables of the constant terminal rate solution of the radial diffusivity equation for transient state condition in terms of dimensionless pressure function. (CO1)

2. (a) Illustrate the constant terminal rate solution for the flow of a real gas.

(CO1/CO2)

OR

- (b) A well is tested by producing it at a constant rate of 1500 stb/d for a period of 100 hours. It is suspected, from seismic and geological evidence, that the well is draining an isolated reservoir block which has approximately a 2 : 1 rectangular geometrical shape and the extended

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drawdown test is intended to confirm this. The reservoir data and flowing bottom hole pressures recorded during the test are detailed below and in table :

$h = 25\text{ft}$, $c = 15 \times 10^{-6} / \text{psi}$, Porosity = .20 B_o 1.25 rb/stb, $r_w = .43 \text{ ft}$, Viscosity = 1 CP

Flowing time (hours)	P_{wf} (psia)	Flowing time (hours)	P_{wf} (psia)
0	3500 (p_i)	20	2762
1	2917	30	2703
2	2900	40	2650
3	2888	50	2597
4	2879	60	2545
5	2869	70	2495
7.5	2848	80	2443
10	2830	90	2392
15	2794	100	2341

(i) Calculate the effective permeability of well.

(ii) Calculate skin factor of the well. (CO1/CO2)

3. (a) Express t_D , in field units with the real time in hours and days and using dimensional analysis, show that both t_D and p_D are dimensionless.

(CO1/CO2)

OR

- (b) What is general theory of well testing ? Explain.

(CO1/CO2)

(3)

4. (a) Explain the concept of constant terminal rate solution for transient state. (CO1)

OR

- (b) Determine the pressure response at the wellbore due to the production of a well which is situated at the centre of a square drainage area with sides measuring : (CO1/CO2)

(i) $L = 100$ ft

(ii) $L = 500$ ft

The relevant reservoir and fluid properties are as follows :

$k = 50$ mD, Porosity = 0.3, Viscosity = 1cp, $c = 15 \times 10^{-6}$ /psi,
 $r_w = 0.3$ ft, CA = 30.9

5. (a) Illustrate basic equation of radial flow of a single-phase fluid and basic differential equation for radial fluid flow. (CO1/CO2)

OR

- (b) Using diffusivity equation, prove dimensionless pressure. (CO1/CO2)

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B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
PETROLEUM REFINING ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What are the basic steps for the crude oil refining ? Classify the different physical and chemical processes involved in the refining of crude oil refining. (CO1)

OR

- (b) Discuss the classification of crude oils. What is the composition of crude oil ? Discuss in detail. (CO1)
2. (a) On which laboratory tests basis the crude oil characterization is done ? Explain on the basis of volatility combustion and viscosity in details. (CO2)

OR

- (b) Explain petroleum refining, operation and optimization. Discuss the selection of processes for optimization. (CO2)

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3. (a) Define the following : (CO2)
- (i) Octane number
 - (ii) Cetane number
 - (iii) Flash and fire point
 - (iv) Diesel index
 - (v) Viscosity index

OR

- (b) On which basis the classifications of continuous distillations columns are categorized ? Discuss. (CO2)
4. (a) What do you mean by dehydration/desaltation ? Draw the flow sheet diagram of crude oil desalting process and discuss its process description stepwise. (CO1/CO2)

OR

- (b) Discuss the main components of distillation columns. Draw the flow sheet diagram of vacuum distillation of reduced crude oil (VDU). Explain it with the process description. (CO1/CO2)
5. (a) What are the purpose of solvent extraction ? What are the most widely used extraction solvent ? Discuss solvent extraction process with its process description. (CO2)

OR

- (b) Why is solvent dewaxing used ? What are the general steps of solvent dewaxing ? Draw the flow sheet diagram of aromatic solvent extraction unit. (CO2)

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B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
OIL FIELD INSTRUMENTATION AND CONTROL

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What do you mean by the following ? (CO1)
- (i) Instrumentation
 - (ii) Process instrumentation
 - (iii) Transducer
 - (iv) Mechanical Amplifier
 - (v) Hydraulic Amplifier

OR

- (b) Discuss in details about the functional elements of a measurement system. Explain the classification of electrical transducer. (CO1)
2. (a) What do you mean by calibration of an instrument ? What is the calibration procedure and how can it be classified ? Discuss the different types of errors in details. (CO2)

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(2)

OR

- (b) Illustrate working of liquid filled thermometer (class-I type) along with the neat and labeled diagram. (CO2)
3. (a) Discuss classification of temperature sensors based on Mechanical sensor, Electrical sensor and Radiation sensors. (CO2)

OR

- (b) Explain resistance temperature detector (RTD) with their principle and working. (CO2)
4. (a) Explain the following principles of thermoelectric temperature measurements : (CO1/CO2)
- (i) Seebeck effect
 - (ii) Peltier effect
 - (iii) Thompson effect

OR

- (b) Discuss the principle, construction and working, advantages, disadvantages and limitations of solid state temperature sensor. (CO1/CO2)
5. (a) Discuss the classification of pressure measuring instruments. Illustrate the well manometer and enlarged leg manometer. (CO2)

OR

- (b) Discuss electrical pressure transducer. Discuss the principle, working and construction of resistance type transducer i.e. strain gauge. (CO2)